

| Model              | Method                     | Cancer-Myth       |                    | CREPE |       | Medical QA |      |      | General QA |       |
|--------------------|----------------------------|-------------------|--------------------|-------|-------|------------|------|------|------------|-------|
|                    |                            | FPQ ↑             | NFP ↑              | FPQ ↑ | TPQ ↑ | MQ ↑       | PQ ↑ | MB ↑ | MMLU ↑     | GSM ↑ |
| <b>Qwen2.5 7B</b>  | Plain                      | 1.7 <sup>T</sup>  | 100.0 <sup>T</sup> | —     | —     | —          | —    | —    | —          | —     |
|                    | DPO (FPQ + twins)          | 2.6 <sup>T</sup>  | 100.0 <sup>T</sup> | —     | —     | —          | —    | —    | —          | —     |
|                    | SFT (FPQ + twins)          | 29.1 <sup>T</sup> | 45.6 <sup>T</sup>  | —     | —     | —          | —    | —    | —          | —     |
|                    | SFT (same chosen)          | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | DPO (FPQ + NFP)            | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | <b>Feature-curated DPO</b> | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | <b>GEPA + SAE feedback</b> | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | <b>GEPA + SAE + NLA</b>    | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
| <b>Gemma-3 12B</b> | Plain                      | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | SFT (same chosen)          | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | DPO (FPQ + NFP)            | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | <b>Feature-curated DPO</b> | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | <b>GEPA + SAE feedback</b> | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | <b>GEPA + SAE + NLA</b>    | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
| <b>Gemma-2 9B</b>  | Plain                      | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | SFT (same chosen)          | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | DPO (FPQ + NFP)            | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | <b>Feature-curated DPO</b> | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | <b>GEPA + SAE feedback</b> | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |
|                    | <b>GEPA + SAE + AO</b>     | —                 | —                  | —     | —     | —          | —    | —    | —          | —     |

**Table S1. Training and internal-description ablations**

Stored results; — means unavailable or not comparable. Values are percentages, one run.  
FPQ / NFP / TPQ: Well ≥ 4. MQ / PQ / MB: MedQA / PubMedQA / Medbullets.  
T: legacy 234 FPQ / 149 natural NFP; twins are synthetic training data, not test NFP.  
Planned same-chosen SFT / natural-NFP DPO / feature feedback remain unmeasured.  
Model versions, counts, judge/protocol differences and source hashes: result\_sources.md.